Study Design: Directed Acyclic Graph and Analytical Framework

Sociodemographic & Lifestyle Factors

Age, Sex, Smoking, Alcohol, Substance Use

Adiposity / Mediators

- BMI

Immuno-Metabolic Markers

CRP (mg/L)

GGT (U/L)

UA (µmol/L)

Clinical Psychiatric & Somatic Outcomes

- Depressive Disorder
- Cardiovascular Disease
- Suicide Attempts
- Judicial History / Criminal Justice

Confounding & Mediation Pathways

Iatrogenic & Clinical Co-factors

- Antipsychotic Metabolic Liability (Olanzapine)
- Hepatic Enzyme Inducers (Carbamazepine, Phenobarbital)
- Thiazide Diuretics (Coaprovel / HCTZ)
- Illness Duration & Renal Function (eGFR)

Analytical Strategy

1 Data Preparation

- ✓ Continuous biomarker scaling
- ✓ Log-transforms
- ✓ Sex-standardization
- ✓ Per-SD effects

2 Regression Analysis

Modified Poisson Regression
(Prevalence Ratios)
for common outcomes
(Depression 41%, CVD 29%)

Firth Penalized Logistic Regression
(aOR) for sparse outcomes
(Suicide attempt 24%, Judicial history 14.5%)

3 Latent Clustering

Gaussian Mixture Models
(BIC selection, bootstrap stability)

Density

Biomarker level

4 Interpretation & Validation

- ✓ Causal inference framework
- ✓ Robustness & sensitivity analyses
- ✓ Clinical implications

Integrative Approach to Understand Immuno-Metabolic Confounding in Schizophrenia